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Introduction

OpenF1 is an open-source API providing detailed Formula 1 telemetry, timing, and session data in JSON and CSV formats.

Historical data (from 2023 onwards) is free and accessible without authentication. Real-time data requires a paid subscription.

You can query the API directly through your browser or HTTP client. Explore the endpoints below to get started.

OpenF1 is an unofficial project and is not associated in any way with the Formula 1 companies. F1, FORMULA ONE, FORMULA 1, FIA FORMULA ONE WORLD CHAMPIONSHIP, GRAND PRIX and related marks are trade marks of Formula One Licensing B.V.

API endpoints

Car data

Some data about each car, at a sample rate of about 3.7 Hz.

```
curl "https://api.openf1.org/v1/car_data?driver_number=55&session_key=9159&speed>=315"
```

Output:


```
[
  {
    "brake": 0,
    "date": "2023-09-15T13:08:19.923000+00:00",
    "driver_number": 55,
    "drs": 12,
    "meeting_key": 1219,
    "n_gear": 8,
    "rpm": 11141,
    "session_key": 9159,
    "speed": 315,
    "throttle": 99
  },
  {
```

```
    "brake": 100,
    "date": "2023-09-15T13:35:41.808000+00:00",
    "driver_number": 55,
    "drs": 8,
    "meeting_key": 1219,
    "n_gear": 8,
    "rpm": 11023,
    "session_key": 9159,
    "speed": 315,
    "throttle": 57
  }
]
```


SAMPLE URL

https://api.openf1.org/v1/car_data?driver_number=55&session_key=9159&speed>=315

ATTRIBUTES

Name	Description
brake	Whether the brake pedal is pressed (`100`) or not (`0`).
date	The UTC date and time, in ISO 8601 format.
driver_number	The unique number assigned to an F1 driver for the season (cf. Wikipedia).
drs	The Drag Reduction System (DRS) status (see mapping table below).
meeting_key	The unique identifier for the meeting. Use `latest` to identify the latest or current meeting.
n_gear	Current gear selection, ranging from 1 to 8. `0` indicates neutral or no gear engaged.
rpm	Revolutions per minute of the engine.
session_key	The unique identifier for the session. Use `latest` to identify the latest or current session.
speed	Velocity of the car in km/h.
throttle	Percentage of maximum engine power being used.

Below is a table that correlates DRS values to its supposed interpretation (from FastF1).

DRS value	Interpretation
0	DRS off
1	DRS off
2	?
3	?
8	Detected, eligible once in activation zone
9	?
10	DRS on
12	DRS on
14	DRS on

Drivers championship (beta)

Provides championship standings for drivers. Only available for race sessions.

```
curl "https://api.openf1.org/v1/championship_drivers?session_key=9839&driver_number=4&driver_number=81"
```

Output:


```
[
  {
    "driver_number": 4,
    "meeting_key": 1276,
    "points_current": 423,
    "points_start": 408,
    "position_current": 1,
    "position_start": 1,
    "session_key": 9839
  },
  {
    "driver_number": 81,
    "meeting_key": 1276,
    "points_current": 410,
    "points_start": 392,
    "position_current": 3,
```

```
    "position_start": 3,
    "session_key": 9839
  }
]
```

SAMPLE URL

https://api.openf1.org/v1/championship_drivers?session_key=9839&driver_number=4&driver_number=81

ATTRIBUTES

Name	Description
driver_number	The unique number assigned to an F1 driver for the season (cf. Wikipedia).
meeting_key	The unique identifier for the meeting. Use `latest` to identify the latest or current meeting.
points_current	Championship points during/after the race (depends on call timing).
points_start	Championship points before the race started.
position_current	Championship position during/after the race (depends on call timing).
position_start	Championship position before the race started.
session_key	The unique identifier for the session. Use `latest` to identify the latest or current session.

Teams championship (beta)

Provides championship standings for teams. Only available for race sessions.

```
curl "https://api.openf1.org/v1/championship_teams?session_key=9839&team_name=McLaren"
```

Output:

```
[
  {
    "meeting_key": 1276,
    "points_current": 833,
    "points_start": 800,
    "position_current": 1,
    "position_start": 1,
    "session_key": 9839,
    "team_name": "McLaren"
  }
]
```

SAMPLE URL

https://api.openf1.org/v1/championship_teams?session_key=9839&team_name=McLaren

ATTRIBUTES

Name	Description
meeting_key	The unique identifier for the meeting. Use `latest` to identify the latest or current meeting.
points_current	Championship points during/after the race (depends on call timing).
points_start	Championship points before the race started.
position_current	Championship position during/after the race (depends on call timing).
position_start	Championship position before the race started.
session_key	The unique identifier for the session. Use `latest` to identify the latest or current session.
team_name	The name of the team.

Drivers

Retrieve detailed information about the drivers participating in a specific session.

```
curl "https://api.openf1.org/v1/drivers?driver_number=1&session_key=9158"
```

Output:

```
[
  {
    "broadcast_name": "M VERSTAPPEN",
    "driver_number": 1,
    "first_name": "Max",
    "full_name": "Max VERSTAPPEN",
    "headshot_url": "https://www.formula1.com/content/dam/fom-website/drivers/M/MAXVER01_Max_Verstappen/maxver01.png.transform/1col/image.png",
    "last_name": "Verstappen",
```

```
    "meeting_key": 1219,
    "name_acronym": "VER",
    "session_key": 9158,
    "team_colour": "3671C6",
    "team_name": "Red Bull Racing"
  }
]
```


SAMPLE URL

https://api.openf1.org/v1/drivers?driver_number=1&session_key=9158

ATTRIBUTES

Name	Description
broadcast_name	The driver's name, as displayed on TV.
country_code (deprecated)	A code that uniquely identifies the country. This field will be removed at the end of the 2026 season.
driver_number	The unique number assigned to an F1 driver for the season (cf. Wikipedia).
first_name	The driver's first name.
full_name	The driver's full name.
headshot_url	URL of the driver's face photo.
last_name	The driver's last name.
meeting_key	The unique identifier for the meeting. Use `latest` to identify the latest or current meeting.
name_acronym	Three-letter acronym of the driver's name.
session_key	The unique identifier for the session. Use `latest` to identify the latest or current session.
team_colour	The hexadecimal color value (RRGGBB) of the driver's team.
team_name	Name of the driver's team.

Intervals

Fetches real-time interval data between drivers and their gap to the race leader. Available during races only, with updates approximately every 4 seconds.

```
curl "https://api.openf1.org/v1/intervals?session_key=9165&interval>0&interval<0.005"
```

Output:


```
[
  {
    "date": "2023-09-17T13:31:02.395000+00:00",
    "driver_number": 1,
    "gap_to_leader": 41.019,
    "interval": 0.003,
    "meeting_key": 1219,
    "session_key": 9165
  }
]
```


SAMPLE URL

https://api.openf1.org/v1/intervals?session_key=9165&interval<0.005

ATTRIBUTES

Name	Description
date	The UTC date and time, in ISO 8601 format.
driver_number	The unique number assigned to an F1 driver for the season (cf. Wikipedia).
gap_to_leader	The time gap to the race leader in seconds, `+1 LAP` if lapped, or `null` for the race leader.
interval	The time gap to the car ahead in seconds, `+1 LAP` if lapped, or `null` for the race leader.
meeting_key	The unique identifier for the meeting. Use `latest` to identify the latest or current meeting.
session_key	The unique identifier for the session. Use `latest` to identify the latest or current session.

Laps

Provides detailed information about individual laps.

```
curl "https://api.openf1.org/v1/laps?session_key=9161&driver_number=63&lap_number=8"
```

Output:


```
[
  {
    "date_start": "2023-09-16T13:59:07.606000+00:00",
    "driver_number": 63,
    "duration_sector_1": 26.966,
    "duration_sector_2": 38.657,
    "duration_sector_3": 26.12,
    "i1_speed": 307,
    "i2_speed": 277,
    "is_pit_out_lap": false,
    "lap_duration": 91.743,
    "lap_number": 8,
    "meeting_key": 1219,
    "segments_sector_1": [2049, 2049, 2049, 2051, 2049, 2051, 2049, 2049],
    "segments_sector_2": [2049, 2049, 2049, 2049, 2049, 2049, 2049, 2049],
    "segments_sector_3": [2048, 2048, 2048, 2048, 2048, 2064, 2064, 2064],
    "session_key": 9161,
    "st_speed": 298
  }
]
```


SAMPLE URL

https://api.openf1.org/v1/laps?session_key=9161&driver_number=63&lap_number=8

ATTRIBUTES

Name	Description
date_start	The UTC starting date and time, in ISO 8601 format. This date is approximate.
driver_number	The unique number assigned to an F1 driver for the season (cf. Wikipedia).
duration_sector_1	The time taken, in seconds, to complete the first sector of the lap.
duration_sector_2	The time taken, in seconds, to complete the second sector of the lap.
duration_sector_3	The time taken, in seconds, to complete the third sector of the lap.
i1_speed	The speed of the car, in km/h, at the first intermediate point on the track.
i2_speed	The speed of the car, in km/h, at the second intermediate point on the track.
is_pit_out_lap	A boolean value indicating whether the lap is an "out lap" from the pit (`true` if it is, `false` otherwise).
lap_duration	The total time taken, in seconds, to complete the entire lap.
lap_number	The sequential number of the lap within the session (starts at 1).
meeting_key	The unique identifier for the meeting. Use `latest` to identify the latest or current meeting.
segments_sector_1	A list of values representing the "mini-sectors" within the first sector (see mapping table below).
segments_sector_2	A list of values representing the "mini-sectors" within the second sector (see mapping table below).
segments_sector_3	A list of values representing the "mini-sectors" within the third sector (see mapping table below).
session_key	The unique identifier for the session. Use `latest` to identify the latest or current session.
st_speed	The speed of the car, in km/h, at the speed trap, which is a specific point on the track where the highest speeds are usually recorded.

Below is a table that correlates segment values to their meaning.

Value	Color
0	not available
2048	yellow sector
2049	green sector
2050	?
2051	purple sector
2052	?
2064	pitlane
2068	?

Segments are not available during races. Also, The segment values may not always align perfectly with the colors shown on TV, for unknown reasons.

Location

The approximate location of the cars on the circuit, at a sample rate of about 3.7 Hz. Useful for gauging their progress along the track, but lacks details about lateral placement — i.e. whether the car is on the left or right side of the track. The origin point (0, 0, 0) appears to be arbitrary and not tied to any specific location on the track.

```
curl "https://api.openf1.org/v1/location?session_key=9161&driver_number=81&date>2023-09-16T13:03:35.200&date<2023-09-16T13:03:35.800"
```

Output:


```
[
  {
    "date": "2023-09-16T13:03:35.292000+00:00",
    "driver_number": 81,
    "meeting_key": 1219,
    "session_key": 9161,
    "x": 567,
    "y": 3195,
    "z": 187
  },
  {
    "date": "2023-09-16T13:03:35.752000+00:00",
    "driver_number": 81,
    "meeting_key": 1219,
    "session_key": 9161,
    "x": 489,
    "y": 3403,
    "z": 186
  }
]
```


SAMPLE URL

https://api.openf1.org/v1/location?session_key=9161&driver_number=81&date>2023-09-16T13:03:35.200&date<2023-09-16T13:03:35.800

ATTRIBUTES

Name	Description
date	The UTC date and time, in ISO 8601 format.
driver_number	The unique number assigned to an F1 driver for the season (cf. Wikipedia).
meeting_key	The unique identifier for the meeting. Use `latest` to identify the latest or current meeting.
session_key	The unique identifier for the session. Use `latest` to identify the latest or current session.
x	The 'x' value in a 3D Cartesian coordinate system representing the current approximate location of the car on the track.
y	The 'y' value in a 3D Cartesian coordinate system representing the current approximate location of the car on the track.
z	The 'z' value in a 3D Cartesian coordinate system representing the current approximate location of the car on the track.

Meetings

Provides information about meetings. A meeting refers to a Grand Prix or testing weekend and usually includes multiple sessions (practice, qualifying, race, ...). Meetings are updated every day at midnight UTC.

```
curl "https://api.openf1.org/v1/meetings?year=2026&country_name=Singapore"
```

Output:


```
[
  {
    "circuit_key": 61,
    "circuit_info_url": "https://api.multiviewer.app/api/v1/circuits/61/2026",
    "circuit_image": "https://media.formula1.com/content/dam/fom-website/2018-redesign-assets/Track%20icons%204x3/Singapore%20carbon.png",
    "circuit_short_name": "Singapore",
    "circuit_type": "Temporary - Street",
    "country_code": "SGP",
    "country_flag": "https://media.formula1.com/content/dam/fom-website/2018-redesign-assets/Flags%2016x9/singapore-flag.png",
    "country_key": 157,
    "country_name": "Singapore",
    "date_end": "2026-10-11T14:00:00+00:00",
    "date_start": "2026-10-09T09:30:00+00:00",
    "gmt_offset": "08:00:00",
    "location": "Marina Bay",
    "meeting_key": 1296,
    "meeting_name": "Singapore Grand Prix",
    "meeting_official_name": "FORMULA 1 SINGAPORE AIRLINES SINGAPORE GRAND PRIX 2026",
    "year": 2026
  }
]
```


SAMPLE URL

https://api.openf1.org/v1/meetings?year=2026&country_name=Singapore

ATTRIBUTES

Name	Description
circuit_key	The unique identifier for the circuit where the event takes place.

Name	Description
circuit_image	An image of the circuit.
circuit_info_url	A URL to a JSON containing detailed circuit info. See FastF1 documentation for details. Data provided by MultiViewer.
circuit_short_name	The short or common name of the circuit where the event takes place.
circuit_type	The type of the circuit ("Permanent", "Temporary - Street", or "Temporary - Road")
country_code	A code that uniquely identifies the country.
country_flag	An image of the country flag.
country_key	The unique identifier for the country where the event takes place.
country_name	The full name of the country where the event takes place.
date_end	The UTC ending date and time, in ISO 8601 format.
date_start	The UTC starting date and time, in ISO 8601 format.
gmt_offset	The difference in hours and minutes between local time at the location of the event and Greenwich Mean Time (GMT).
location	The city or geographical location where the event takes place.
meeting_key	The unique identifier for the meeting. Use `latest` to identify the latest or current meeting.
meeting_name	The name of the meeting.
meeting_official_name	The official name of the meeting.
year	The year the event takes place.

Overtakes

Provides information about overtakes. An overtake refers to one driver (the overtaking driver) exchanging positions with another driver (the overtaken driver). This includes both on-track passes and position changes resulting from pit stops or post-race penalties. This data is only available during races and may be incomplete.

```
curl "https://api.openf1.org/v1/overtakes?session_key=9636&overtaking_driver_number=63&overtaken_driver_number=4&position=1"
```

Output:


```
[
  {
    "date": "2024-11-03T15:50:07.565000+00:00",
    "meeting_key": 1249,
    "overtaken_driver_number": 4,
    "overtaking_driver_number": 63,
    "position": 1,
    "session_key": 9636
  }
]
```


SAMPLE URL

https://api.openf1.org/v1/overtakes?session_key=9636&overtaking_driver_number=63&overtaken_driver_number=4&position=1

ATTRIBUTES

Name	Description
date	The UTC date and time, in ISO 8601 format.
meeting_key	The unique identifier for the meeting. Use `latest` to identify the latest or current meeting.
overtaken_driver_number	The unique number assigned to the overtaken F1 driver (cf. Wikipedia).
overtaking_driver_number	The unique number assigned to the overtaking F1 driver (cf. Wikipedia).
position	The position of the overtaking F1 driver after the overtake was completed (starts at 1).
session_key	The unique identifier for the session. Use `latest` to identify the latest or current session.

Pit

Provides information about cars going through the pit lane.

```
curl "https://api.openf1.org/v1/pit?session_key=9877&stop_duration<2.3"
```

Output:

```
[
  {
    "date": "2025-10-26T20:46:37.358000+00:00",
    "driver_number": 16,
    "lane_duration": 22.215,
    "lap_number": 31,
    "meeting_key": 1272,
    "pit_duration": 22.215,
    "session_key": 9877,
    "stop_duration": 2.2
  },
  {
    "date": "2025-10-26T21:09:49.689000+00:00",
    "driver_number": 81,
    "lane_duration": 22.159,
    "lap_number": 47,
    "meeting_key": 1272,
    "pit_duration": 22.159,
    "session_key": 9877,
    "stop_duration": 2.1
  }
]
```


SAMPLE URL

https://api.openf1.org/v1/pit?session_key=9877&stop_duration<2.3

ATTRIBUTES

Name	Description
date	The UTC date and time, in ISO 8601 format.
driver_number	The unique number assigned to an F1 driver for the season (cf. Wikipedia).
lane_duration	The time spent in the pit lane, in seconds.
lap_number	The sequential number of the lap within the session (starts at 1).
meeting_key	The unique identifier for the meeting. Use `latest` to identify the latest or current meeting.
pit_duration (deprecated)	Same as 'lane_duration'. This field will be removed at the end of the 2026 season.
session_key	The unique identifier for the session. Use `latest` to identify the latest or current session.
stop_duration	The stationary pit stop time, in seconds. This field is only available from the 2024 US GP onwards.

Position

Provides driver positions throughout a session, including initial placement and subsequent changes.

```
curl "https://api.openf1.org/v1/position?meeting_key=1217&driver_number=40&position<=3"
```

Output:


```
[
  {
    "date": "2023-08-26T09:30:47.199000+00:00",
    "driver_number": 40,
    "meeting_key": 1217,
    "position": 2,
    "session_key": 9144
  },
  {
    "date": "2023-08-26T09:35:51.477000+00:00",
    "driver_number": 40,
    "meeting_key": 1217,
    "position": 3,
    "session_key": 9144
  }
]
```


SAMPLE URL

https://api.openf1.org/v1/position?meeting_key=1217&driver_number=40&position<=3

ATTRIBUTES

Name	Description
date	The UTC date and time, in ISO 8601 format.
driver_number	The unique number assigned to an F1 driver for the season (cf. Wikipedia).
meeting_key	The unique identifier for the meeting. Use `latest` to identify the latest or current meeting.
position	Position of the driver (starts at 1).
session_key	The unique identifier for the session. Use `latest` to identify the latest or current session.

Race control

Provides information about race control (session status, racing incidents, flags, safety car, ...).

```
curl "https://api.openf1.org/v1/race_control?flag=BLACK AND WHITE&driver_number=1&date>=2023-01-01&date<2023-09-01"
```

Output:


```
[
 {
   "category": "Flag",
   "date": "2023-06-04T14:21:01+00:00",
   "driver_number": 1,
   "flag": "BLACK AND WHITE",
   "lap_number": 59,
   "meeting_key": 1211,
   "message": "BLACK AND WHITE FLAG FOR CAR 1 (VER) - TRACK LIMITS",
   "qualifying_phase": null,
   "scope": "Driver",
   "sector": null,
   "session_key": 9102
 }
]
```


SAMPLE URL

https://api.openf1.org/v1/race_control?flag=BLACK AND WHITE&driver_number=1&date>=2023-01-01&date<2023-09-01

ATTRIBUTES

Name	Description
category	The category of the event (`SessionStatus`, `CarEvent`, `Drs`, `Flag`, `SafetyCar`, ...).
date	The UTC date and time, in ISO 8601 format.
driver_number	The unique number assigned to an F1 driver for the season (cf. Wikipedia).
flag	Type of flag displayed (`GREEN`, `YELLOW`, `DOUBLE YELLOW`, `CHEQUERED`, ...).
lap_number	The sequential number of the lap within the session (starts at 1), in a race.
meeting_key	The unique identifier for the meeting. Use `latest` to identify the latest or current meeting.
message	Description of the event or action.
qualifying_phase	The specific phase (`1`, `2`, or `3`) if the session is a qualifying session.
scope	The scope of the event (`Track`, `Driver`, `Sector`, ...).
sector	Segment ("mini-sector") of the track where the event occurred? (starts at 1).
session_key	The unique identifier for the session. Use `latest` to identify the latest or current session.

Sessions

Provides information about sessions. A session refers to a distinct period of track activity during a Grand Prix or testing weekend (practice, qualifying, sprint, race, ...). Sessions are updated every day at midnight UTC.

```
curl "https://api.openf1.org/v1/sessions?country_name=Belgium&session_name=Sprint%20Qualifying&year=2023"
```

Output:


```
[
 {
   "circuit_key": 7,
   "circuit_short_name": "Spa-Francorchamps",
   "country_code": "BEL",
   "country_key": 16,
   "country_name": "Belgium",
   "date_end": "2023-07-29T15:35:00+00:00",
   "date_start": "2023-07-29T15:05:00+00:00",
   "gmt_offset": "02:00:00",
   "location": "Spa-Francorchamps",
   "meeting_key": 1216,
   "session_key": 9140,
   "session_name": "Sprint Qualifying",
   "session_type": "Sprint Qualifying",
   "year": 2023
 }
]
```


SAMPLE URL

https://api.openf1.org/v1/sessions?country_name=Belgium&session_name=Sprint%20Qualifying&year=2023

ATTRIBUTES

Name	Description
circuit_key	The unique identifier for the circuit where the event takes place.
circuit_short_name	The short or common name of the circuit where the event takes place.
country_code	A code that uniquely identifies the country.
country_key	The unique identifier for the country where the event takes place.
country_name	The full name of the country where the event takes place.
date_end	The UTC ending date and time, in ISO 8601 format.
date_start	The UTC starting date and time, in ISO 8601 format.
gmt_offset	The difference in hours and minutes between local time at the location of the event and Greenwich Mean Time (GMT).
location	The city or geographical location where the event takes place.
meeting_key	The unique identifier for the meeting. Use `latest` to identify the latest or current meeting.
session_key	The unique identifier for the session. Use `latest` to identify the latest or current session.
session_name	The name of the session (`Practice 1`, `Qualifying`, `Race`, ...).
session_type	The type of the session (`Practice`, `Qualifying`, `Race`, ...).
year	The year the event takes place.

Session result

Provides standings after a session. This data becomes available a few minutes after the official results are published on the official Formula 1 website.

```
curl "https://api.openf1.org/v1/session_result?session_key=7782&position%3C=3"
```

Output:


```
[
  {
    "dnf": false,
    "dns": false,
    "dsq": false,
    "driver_number": 1,
    "duration": 77.565,
    "gap_to_leader": 0,
    "number_of_laps": 24,
    "meeting_key": 1143,
    "position": 1,
    "session_key": 7782
  },
  {
    "dnf": false,
    "dns": false,
    "dsq": false,
    "driver_number": 14,
    "duration": 77.727,
    "gap_to_leader": 0.162,
    "number_of_laps": 26,
    "meeting_key": 1143,
    "position": 2,
    "session_key": 7782
  },
  {
    "dnf": false,
    "dns": false,
    "dsq": false,
    "driver_number": 31,
    "duration": 77.938,
    "gap_to_leader": 0.373,
    "number_of_laps": 23,
    "meeting_key": 1143,
    "position": 3,
    "session_key": 7782
  }
]
```


SAMPLE URL

https://api.openf1.org/v1/session_result?session_key=7782&position<=3

ATTRIBUTES

Name	Description
dnf	Indicates whether the driver *Did Not Finish* the race. This can be `true` only for qualifying and race sessions.
dns	Indicates whether the driver *Did Not Start* the race. This can be `true` only for qualifying and race sessions.
dsq	Indicates whether the driver was disqualified.

Name	Description
driver_number	The unique number assigned to an F1 driver for the season (cf. Wikipedia).
duration	Either the best lap time (for practice or qualifying), or the total race time (for races), in seconds. In qualifying, this is an array of three values for Q1, Q2, and Q3.
gap_to_leader	The time gap to the session leader in seconds, or `+N LAP(S)` if the driver was lapped. In qualifying, this is an array of three values for Q1, Q2, and Q3.
number_of_laps	Total number of laps completed during the session.
meeting_key	The unique identifier for the meeting. Use `latest` to identify the latest or current meeting.
position	The driver's final position at the end of the session.
session_key	The unique identifier for the session. Use `latest` to identify the latest or current session.

Starting grid

Provides the starting grid for the upcoming race. This data becomes available a few minutes after the official results are published on the official Formula 1 website.

```
curl "https://api.openf1.org/v1/starting_grid?session_key=7783&position%3C=3"
```

Output:


```
[
  {
    "position": 1,
    "driver_number": 1,
    "lap_duration": 76.732,
    "meeting_key": 1143,
    "session_key": 7783
  },
  {
    "position": 2,
    "driver_number": 63,
    "lap_duration": 76.968,
    "meeting_key": 1143,
    "session_key": 7783
  },
  {
    "position": 3,
    "driver_number": 44,
    "lap_duration": 77.104,
    "meeting_key": 1143,
    "session_key": 7783
  }
]
```


SAMPLE URL

https://api.openf1.org/v1/starting_grid?session_key=7783&position<=3

ATTRIBUTES

Name	Description
driver_number	The unique number assigned to an F1 driver for the season (cf. Wikipedia).
lap_duration	Duration, in seconds, of the qualifying lap.
meeting_key	The unique identifier for the meeting. Use `latest` to identify the latest or current meeting.
position	Position on the grid.
session_key	The unique identifier for the session. Use `latest` to identify the latest or current session.

Stints

Provides information about individual stints. A stint refers to a period of continuous driving by a driver during a session.

```
curl "https://api.openf1.org/v1/stints?session_key=9165&tyre_age_at_start>=3"
```

Output:


```
[
  {
    "compound": "SOFT",
    "driver_number": 16,
    "lap_end": 20,
    "lap_start": 1,
    "meeting_key": 1219,
    "session_key": 9165,
    "stint_number": 1,
    "tyre_age_at_start": 3
  },
```

```
  {
    "compound": "SOFT",
    "driver_number": 20,
    "lap_end": 62,
    "lap_start": 44,
    "meeting_key": 1219,
    "session_key": 9165,
    "stint_number": 3,
    "tyre_age_at_start": 3
  }
]
```


SAMPLE URL

https://api.openf1.org/v1/stints?session_key=9165&tyre_age_at_start>=3

ATTRIBUTES

Name	Description
compound	The specific compound of tyre used during the stint (`SOFT`, `MEDIUM`, `HARD`, ...).
driver_number	The unique number assigned to an F1 driver for the season (cf. Wikipedia).
lap_end	Number of the last completed lap in this stint.
lap_start	Number of the initial lap in this stint (starts at 1).
meeting_key	The unique identifier for the meeting. Use `latest` to identify the latest or current meeting.
session_key	The unique identifier for the session. Use `latest` to identify the latest or current session.
stint_number	The sequential number of the stint within the session (starts at 1).
tyre_age_at_start	The age of the tyres at the start of the stint, in laps completed.

Team radio

Provides a collection of radio exchanges between Formula 1 drivers and their respective teams during sessions. Please note that only a limited selection of communications are included, not the complete record of radio interactions.

```
curl "https://api.openf1.org/v1/team_radio?session_key=9158&driver_number=11"
```

Output:


```
[
  {
    "date": "2023-09-15T09:40:43.005000+00:00",
    "driver_number": 11,
    "meeting_key": 1219,
    "recording_url": "https://livetiming.formula1.com/static/2023/2023-09-17_Singapore_Grand_Prix/2023-09-15_Practice_1/TeamRadio/SERPER01_11_20230915_104008.mp3",
    "session_key": 9158
  },
  {
    "date": "2023-09-15T10:32:47.325000+00:00",
    "driver_number": 11,
    "meeting_key": 1219,
    "recording_url": "https://livetiming.formula1.com/static/2023/2023-09-17_Singapore_Grand_Prix/2023-09-15_Practice_1/TeamRadio/SERPER01_11_20230915_113201.mp3",
    "session_key": 9158
  }
]
```


SAMPLE URL

https://api.openf1.org/v1/team_radio?session_key=9158&driver_number=11

ATTRIBUTES

Name	Description
date	The UTC date and time, in ISO 8601 format.
driver_number	The unique number assigned to an F1 driver for the season (cf. Wikipedia).
meeting_key	The unique identifier for the meeting. Use `latest` to identify the latest or current meeting.
recording_url	URL of the radio recording.
session_key	The unique identifier for the session. Use `latest` to identify the latest or current session.

Weather

The weather over the track, updated every minute.

```
curl "https://api.openf1.org/v1/weather?meeting_key=1208&wind_direction>=130&track_temperature>=52"
```

Output:


```
[
  {
    "air_temperature": 27.8,
    "date": "2023-05-07T18:42:25.233000+00:00",
    "humidity": 58,
    "meeting_key": 1208,
    "pressure": 1018.7,
    "rainfall": 0,
    "session_key": 9078,
    "track_temperature": 52.5,
    "wind_direction": 136,
    "wind_speed": 2.4
  }
]
```


SAMPLE URL

https://api.openf1.org/v1/weather?meeting_key=1208&wind_direction>=130&track_temperature>=52

ATTRIBUTES

Name	Description
air_temperature	Air temperature (°C).
date	The UTC date and time, in ISO 8601 format.
humidity	Relative humidity (%).
meeting_key	The unique identifier for the meeting. Use `latest` to identify the latest or current meeting.
pressure	Air pressure (mbar).
rainfall	Whether there is rainfall.
session_key	The unique identifier for the session. Use `latest` to identify the latest or current session.
track_temperature	Track temperature (°C).
wind_direction	Wind direction (°), from 0° to 359°.
wind_speed	Wind speed (m/s).

Data filtering

Refine your query by including parameters directly in the URL.
Results can be filtered by any attribute, except arrays.

Example
To fetch pit-out laps for driver number 55 (Carlos Sainz) that last at least 2 minutes, use: https://api.openf1.org/v1/laps?session_key=9222&driver_number=55&is_pit_out_lap=true&lap_duration>=120

Time-Based Filtering

You can narrow down your results using time ranges.

Example
To get all sessions in September 2023, use: https://api.openf1.org/v1/sessions?date_start>=2023-09-01&date_end<=2023-09-30

The API supports a wide range of date formats (those compatible with Python's `dateutil.parser.parse` method). Examples include:

- "2021-09-10"
- "2021-09-10T14:30:20"
- "2021-09-10T14:30:20+00:00"
- "09/10/2021"
- "09-10-2021"
- "Fri Sep 10 14:30:20 2021"
- "10 September 2021"
- "Sep 10, 2021"
- "2021-09-10 14:30:20 UTC"
- "2021-09-10 14:30:20+00:00"
- "2021-09-10 14:30:20 EST"
- ...and many more.

CSV Format

To receive your query results in CSV format instead of the default JSON, simply append the query parameter `csv=true` to your URL. This feature is particularly handy to import the data into spreadsheet software like Microsoft Excel.

Example
To get all sessions for the year 2023 in CSV format, use the following URL: https://api.openf1.org/v1/sessions?year=2023&csv=true

Tutorials

- Creating an interactive strategy dashboard with Python, by @bordanattila
- Streaming live data with OpenF1

Have a great OpenF1 tutorial to share? We'd love to feature it!

Community and Support

For community-driven questions or feedback, we encourage you to participate in our Github Discussions. If you encounter any bugs or issues, please report them by creating a new issue on our Github repository.

For specialized inquiries, feel free to reach out to us directly. However, for general support questions, please use the Github Discussions platform to ensure that the entire community can benefit from the answers.

Contributing

OpenF1's mission is to democratize Formula 1 data by making it open and accessible to everyone.
We welcome all contributions that help advance this goal!

To get started, please review our contribution guidelines.

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